Lesson Plan · Tuesday, November 10

Algebra I · Mr. Pandey · Module 3, Topic 1, Lesson 2: Literal Equations (Day 2 of 2) · essentials only (✪)

Objectives (students will…)

- Define literal equation.
- Solve formulas for a specified variable and justify the steps.
- Use a rewritten formula to find an unknown measure.

TEKS

- **A.12E** solve formulas and other literal equations for a specified variable
- A.3A, A.2B
- Process: A.1A

Key terms

- literal equation

Materials

- Student Edition pp. 362–367 (Activities 2.2, 2.3, Talk the Talk)
- Calculators (3.14 for π)
- Whiteboards

Bell work (on the board)

- Write $4x + 2y = 10$ in slope-intercept form $y = -2x + 5$
- $3x - 7 = 11$ $x = 6$
- $P = 2\ell + 2w$. Find ℓ when $P = 30$, $w = 5$ $\ell = 10$
- $C = 5/9(F - 32)$. Find C when $F = 50$ $C = 10$

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	Bell work	Q3–4 lead into today: we keep solving the same formula for different letters.
5–25	✪ Activity 2.2: Rewriting Literal Equations · SE pp. 362–364	Define *literal equation*. Q1 solve $C = 5/9(F - 32)$ for F: $F = 9/5\ C + 32$. Q2 77°F vs 30°C: Mateo is warmer: 30°C = 86°F. Q3 error analysis: Ethan didn't distribute 5/9 to the 32; Minh multiplied by 9 wrong (45F, 1440). Worked Example (clear fractions), then Q4: $9C - 5F = -160$.
25–45	✪ Activity 2.3: Solving Literal Equations for Specific Variables · SE pp. 365–366	Q1 $V = Bh \rightarrow h = V/B$; $V = 450$, $B = 75 \rightarrow 6$ in. Q2 cone $V = \frac{1}{3}\pi r^2 h \rightarrow h = 3V/(\pi r^2)$. Q3 trapezoid $A = \frac{1}{2}h(b_1 + b_2) \rightarrow h = 2A/(b_1 + b_2)$; $A = 24$, bases 9 and 7 → 3 cm. Q4 $Z = A/B + C/D$ for A: $A = BZ - BC/D$ (part b, solve for D, is the challenge).
45–57	✪ Talk the Talk: Rearrange It · SE p. 367 (exit ticket)	Independent. 1. $S = 2\pi rh$ for h $h = S/(2\pi r)$ 2. $S = 37.68$, $r = 1.5$ $h = 4$ in. 3. $S = 2\pi r^2 + 2\pi rh$ for h $h = (S - 2\pi r^2)/(2\pi r)$ 4. $S = 51.81$, $r = 1.5$ $h = 4$ in. IEP/504: read aloud on request, properties/notes card, extra time.
57–60	Wrap up	Essential Question (SE p. 357). Homework: Lesson 2 Assignment Practice 2–3 and Prepare (SE p. 370). Prepare previews inequalities.

Check for understanding

- Whiteboards: "Solve $A = \ell w$ for w." $w = A/\ell$
- Thumbs: "To get F alone, subtract 32 first." Down: multiply by 9/5 first
- Activity 2.3 Q1–4 are the TE's check points.

After class

- Missed Talk the Talk 1 or 3 → Skills Practice Set D (literal equations).
- Missed 2 or 4 only → calculator/substitution check, small group.
- Prepare answers: $x = 5$, $x > 7$, $x \leq -3$, $x < -4$.

All of Day 2 is starred. Prepare Q4 ($-x > 4 \rightarrow x < -4$) sets up flipping the sign in Lesson 3.